

Nebulae Smart Metering HES – Scalability Report

Document Date: September, 2023

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Typographic Conventions

This document uses the typographic conventions as shown below:

Visual Cue	Meaning
Bold Type with Initial Capital letters	All headings and Sub headings Titles in a document are displayed in bold type with initial capital letters; Example: General Description, Features.
Bold Type with Italic Letters	All Definitions, Figure and Table Headings are displayed in Italics.
1., 2.	Numbered steps are used in a list of items, when the sequence of items is important. such as steps listed in procedure.
•	Bullets are used in a list of items when the sequence of items is not important.

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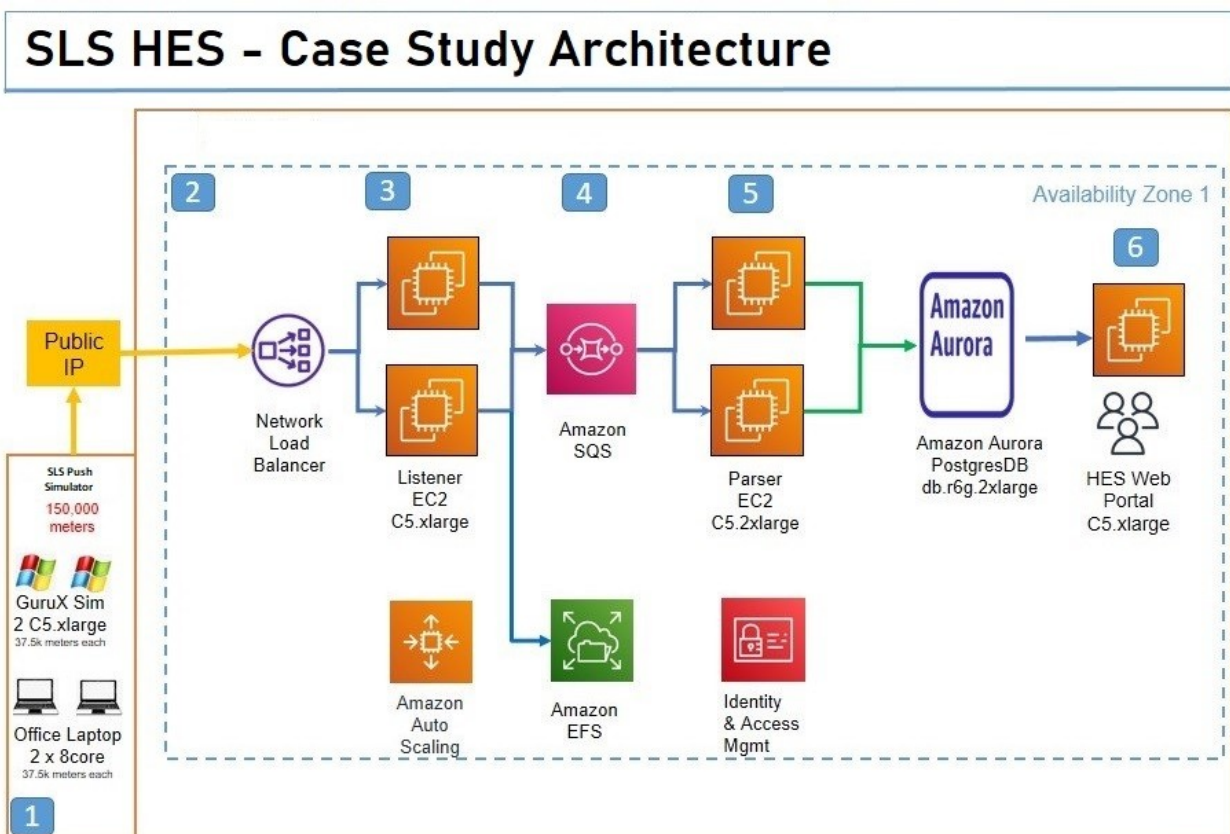
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1 Objective

The objective is to simulate 1.5L meter's 15-minute IP data to be ingested, parsed and inserted for viewing in 15-minutes. This architecture diagram defines key components of the PoC and the necessary process, program and modules involved in achieving this objective.

2 Architecture Details

The architecture leverages horizontal scalability using multiple components to receive/send data from devices. Below is the explanation of the components used:



3 Analysis

1. Meter Population: 150,000
2. Time to Process Data: 15 minutes; Type of Data: IP Data
3. End State: Data from the last 15 min interval available the min after the next 15-min interval. For e.g. Data from a meter from the time period 10:00 to 10:15 is viewable in the web page at 10:31AM

4 Deliverables

Simulated Data: SLS Simulator Push DLMS messages from Four Windows Simulator servers, sending 37.5K each. Multithreaded pooling simulates multiple meters sending data at the same instant to the load balancer, which in turn routes the network traffic to the listeners that are part of its target group.

PoC Cloud Setup: VPC, subnets, security groups, load balancers, target groups, AWS EC2, RDS, SQS, Lambda functions and Eventbridge Events.

Network Load Balancer: Network Load Balancer auto scales as the load increases. The EC2 Listeners were also part of an Auto Scaling group that would spin new instances to meet the load demands. The service would compile the message from one meter store it in EFS and add to the SQS

SQS (standard): SQS will be the source for the pooler code in the Parser that will batch a bunch of messages and process them in batch

Parser Code: Parser code, takes a batch of messages (say 250) and does 1 select for meta data, parses and does one batch insert into PostgreSQL

Data Viewing: Data inserted into PostgreSQL could be viewed in the web application of the HES.